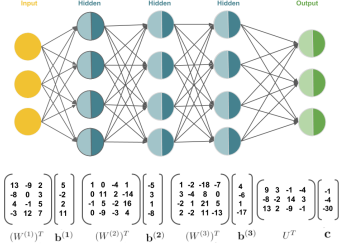
# Introduction to Machine Learning

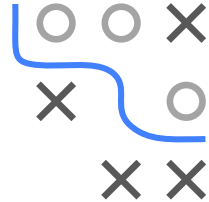
## Neural Networks
## MLP – Multi-Layer Feedforward Neural Networks




**Learning goals**

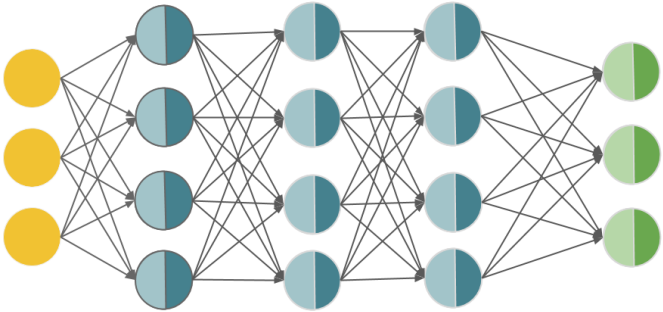
- Architectures of deep neural networks
- Deep neural networks as chained functions

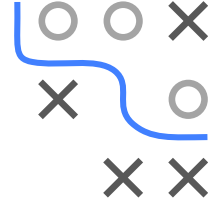
# FEEDFORWARD NEURAL NETWORKS



- We will now extend the model class once again, such that we allow an arbitrary amount $l$ of hidden layers.
- The general term for this model class is (multi-layer) **feedforward networks** (inputs are passed through the network from left to right, no feedback-loops are allowed)

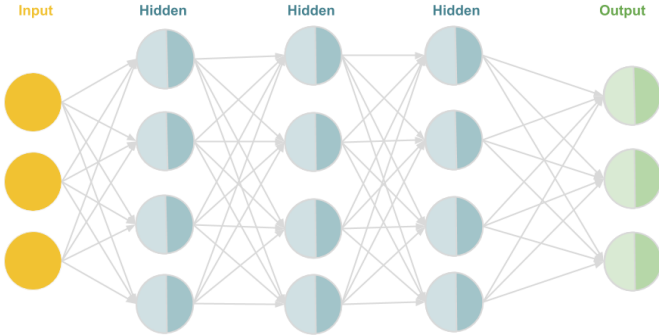
# FEEDFORWARD NEURAL NETWORKS: EXAMPLE



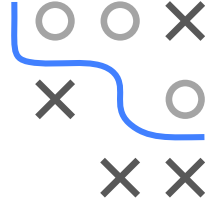



$$\underbrace{\begin{pmatrix} 13 & -9 & 2 \\ -8 & 0 & 3 \\ 4 & -1 & 5 \\ -3 & 12 & 7 \end{pmatrix}}_{(W^{(1)})^T} \underbrace{\begin{pmatrix} 5 \\ -2 \\ 2 \\ 11 \end{pmatrix}}_{\mathbf{b}^{(1)}} \underbrace{\begin{pmatrix} 1 & 0 & -4 & 1 \\ 0 & 11 & 2 & -14 \\ -1 & 5 & -2 & 16 \\ 0 & -9 & -3 & 4 \end{pmatrix}}_{(W^{(2)})^T} \underbrace{\begin{pmatrix} -5 \\ 3 \\ 1 \\ -8 \end{pmatrix}}_{\mathbf{b}^{(2)}} \underbrace{\begin{pmatrix} 1 & -2 & -18 & -7 \\ 3 & -4 & 8 & 0 \\ -2 & 1 & 21 & 5 \\ 2 & -2 & 11 & -13 \end{pmatrix}}_{(W^{(3)})^T} \underbrace{\begin{pmatrix} 4 \\ -6 \\ 1 \\ -17 \end{pmatrix}}_{\mathbf{b}^{(3)}} \underbrace{\begin{pmatrix} 9 & 3 & -1 & -4 \\ -8 & -2 & 14 & 3 \\ 13 & 2 & -9 & -1 \end{pmatrix}}_{U^T} \underbrace{\begin{pmatrix} -1 \\ -4 \\ -30 \end{pmatrix}}_{\mathbf{c}}$$

# FEEDFORWARD NEURAL NETWORKS: EXAMPLE



$\mathbf{x}$



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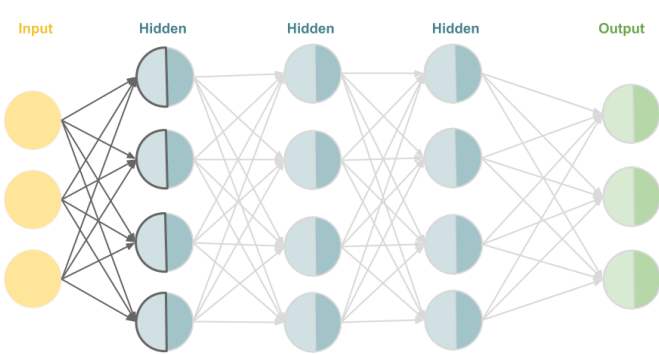
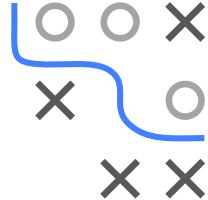
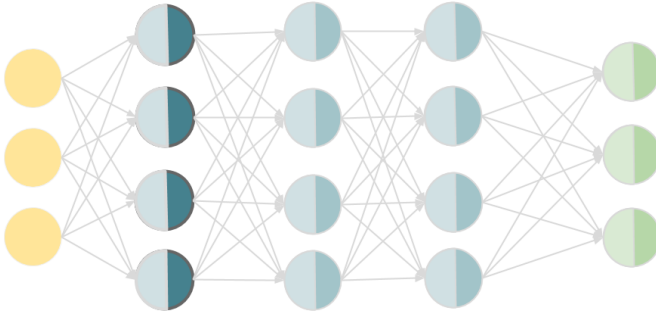


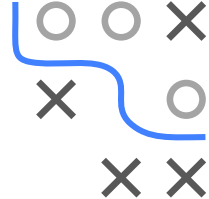



$$\begin{bmatrix} 7 \\ 1 \\ -4 \end{bmatrix} \begin{pmatrix} 7*13 + 1*(-9) + (-4)*2 + 5 \\ 7*(-8) + 1*0 + (-4)*3 + (-2) \\ 7*4 + 1*(-1) + (-4)*5 + 2 \\ 7*(-3) + 1*12 + (-4)*7 + 11 \end{pmatrix}$$

$\mathbf{x}$ $\quad \mathbf{z}_{in}^{(1)} = W^{(1)T}\mathbf{x} + \mathbf{b}^{(1)}$

# FEEDFORWARD NEURAL NETWORKS: EXAMPLE

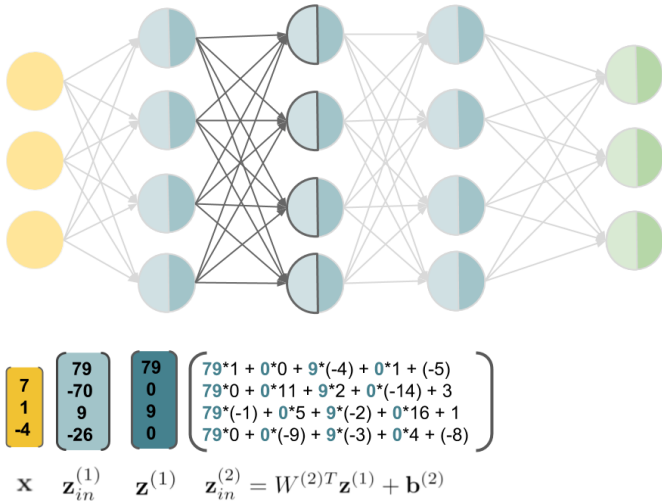


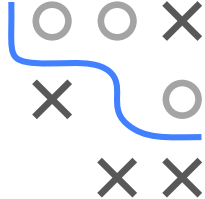



$$\begin{bmatrix} 7 \\ 1 \\ -4 \end{bmatrix} \quad \begin{bmatrix} 79 \\ -70 \\ 9 \\ -26 \end{bmatrix} \quad \begin{pmatrix} \max(0, 79) \\ \max(0, -70) \\ \max(0, 9) \\ \max(0, -26) \end{pmatrix}$$

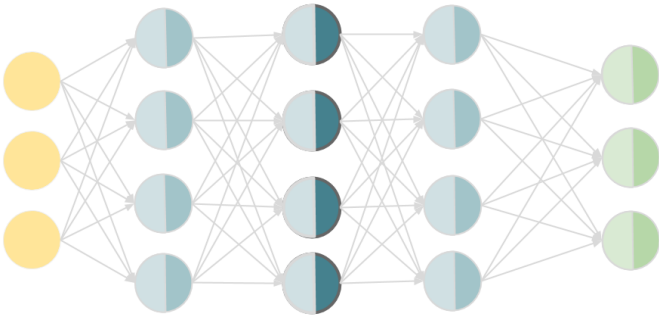
$\mathbf{x}$ $\quad \mathbf{z}_{in}^{(1)}$ $\quad \mathbf{z}^{(1)} = \mathbf{z}_{out}^{(1)} = \sigma(\mathbf{z}_{in}^{(1)})$

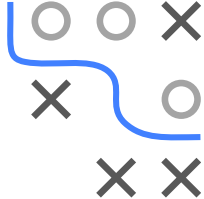
# FEEDFORWARD NEURAL NETWORKS: EXAMPLE



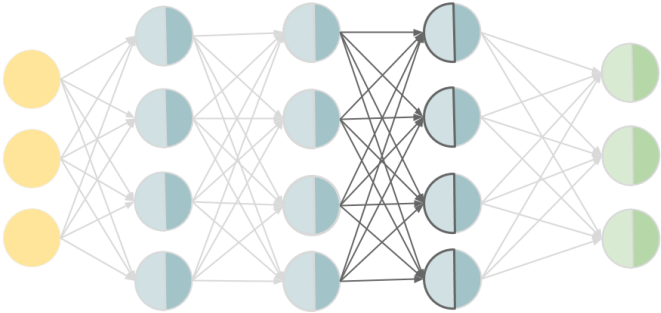
# FEEDFORWARD NEURAL NETWORKS: EXAMPLE

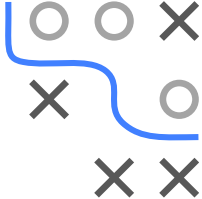


$$\mathbf{x} = \begin{bmatrix} 7 \\ 1 \\ -4 \end{bmatrix} \quad \mathbf{z}_{in}^{(1)} = \begin{bmatrix} 79 \\ -70 \\ 9 \\ -26 \end{bmatrix} \quad \mathbf{z}^{(1)} = \begin{bmatrix} 79 \\ 0 \\ 9 \\ 0 \end{bmatrix} \quad \mathbf{z}_{in}^{(2)} = \begin{bmatrix} 38 \\ 21 \\ -96 \\ -35 \end{bmatrix} \quad \mathbf{z}^{(2)} = \mathbf{z}_{out}^{(2)} = \sigma(\mathbf{z}_{in}^{(2)}) = \begin{pmatrix} \max(0, 38) \\ \max(0, 21) \\ \max(0, -96) \\ \max(0, -36) \end{pmatrix}$$

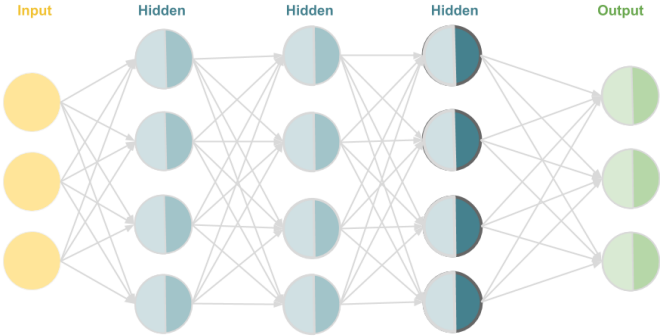
# FEEDFORWARD NEURAL NETWORKS: EXAMPLE

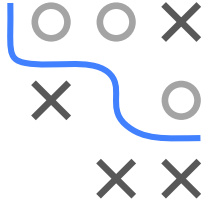


| | | | | | |
|---|---|---|---|---|---|
| 7 | 79 | 79 | 38 | 38 | 38*1 + 21*(-2) + 0*(-18) + 0*(-7) + 4 |
| 1 | -70 | 0 | 21 | 21 | 38*3 + 21*(-4) + 0*8 + 0*0 + (-6) |
| -4 | 9 | 9 | -96 | 0 | 38*(-2) + 21*1 + 0*21 + 0*5 + 1 |
| | -26 | 0 | -35 | 0 | 38*2 + 21*(-2) + 0*11 + 0*(-13) + (-17) |
| $\mathbf{x}$ | $\mathbf{z}_{in}^{(1)}$ | $\mathbf{z}^{(1)}$ | $\mathbf{z}_{in}^{(2)}$ | $\mathbf{z}^{(2)}$ | $\mathbf{z}_{in}^{(3)} = W^{(3)T}\mathbf{z}^{(2)} + \mathbf{b}^{(3)}$ |

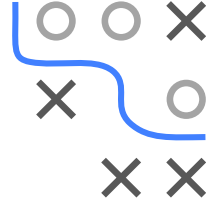
# FEEDFORWARD NEURAL NETWORKS: EXAMPLE



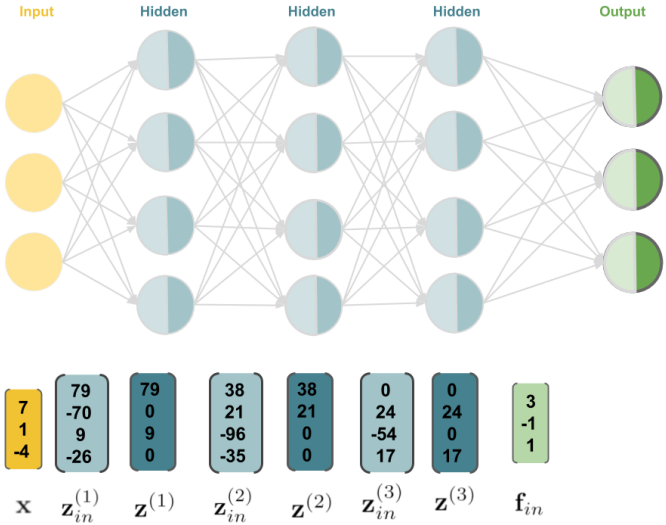
| | | | | | | |
|---|---|---|---|---|---|---|
| 7<br>1<br>-4 | 79<br>-70<br>9<br>-26 | 79<br>0<br>9<br>0 | 38<br>21<br>-96<br>-35 | 38<br>21<br>0<br>0 | 0<br>24<br>-54<br>17 | max(0, 0)<br>max(0, 24)<br>max(0, -54)<br>max(0, 17) |
| $\mathbf{x}$ | $\mathbf{z}_{in}^{(1)}$ | $\mathbf{z}^{(1)}$ | $\mathbf{z}_{in}^{(2)}$ | $\mathbf{z}^{(2)}$ | $\mathbf{z}_{in}^{(3)}$ | $\mathbf{z}^{(3)} = \mathbf{z}_{out}^{(3)} = \sigma(\mathbf{z}_{in}^{(3)})$ |

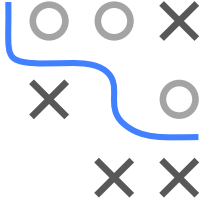
# FEEDFORWARD NEURAL NETWORKS: EXAMPLE



| $\mathbf{x}$ | $\mathbf{z}_{in}^{(1)}$ | $\mathbf{z}^{(1)}$ | $\mathbf{z}_{in}^{(2)}$ | $\mathbf{z}^{(2)}$ | $\mathbf{z}_{in}^{(3)}$ | $\mathbf{z}^{(3)}$ | $\mathbf{f}_{in} = U^T\mathbf{z}^{(3)} + \mathbf{c}$ |
|---|---|---|---|---|---|---|---|
| 7 | 79 | 79 | 38 | 38 | 0 | 0 | 24*3 + 17*(-4) + (-1) |
| 1 | -70 | 0 | 21 | 21 | 24 | 24 | 24*(-2) + 17*3 + (-4) |
| -4 | 9 | 9 | -96 | 0 | -54 | 0 | 24*2 + 17*(-1) + (-30) |
| | -26 | 0 | -35 | 0 | 17 | 17 | |

# FEEDFORWARD NEURAL NETWORKS: EXAMPLE



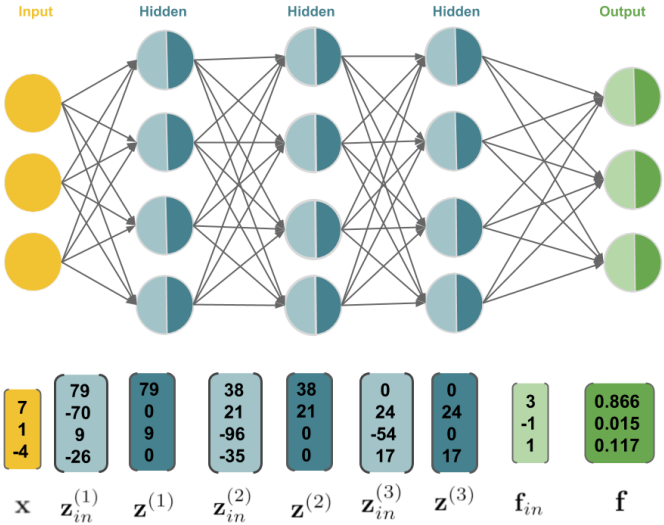



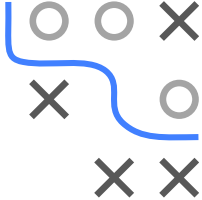
# FEEDFORWARD NEURAL NETWORKS: EXAMPLE

$$\mathbf{f}_{in} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

$$\mathbf{f} = \mathbf{f}_{out} = \tau(\mathbf{f}_{in}) = \begin{pmatrix} \frac{\exp(3)}{\exp 3 + \exp(-1) + \exp(1)} \\ \frac{\exp(-1)}{\exp 3 + \exp(-1) + \exp(1)} \\ \frac{\exp(1)}{\exp 3 + \exp(-1) + \exp(1)} \end{pmatrix}$$

# FEEDFORWARD NEURAL NETWORKS: EXAMPLE



# WHY ADD MORE LAYERS?

- Multiple layers allow for the extraction of more and more abstract representations.
- Each layer in a feed-forward neural network adds its own degree of non-linearity to the model.

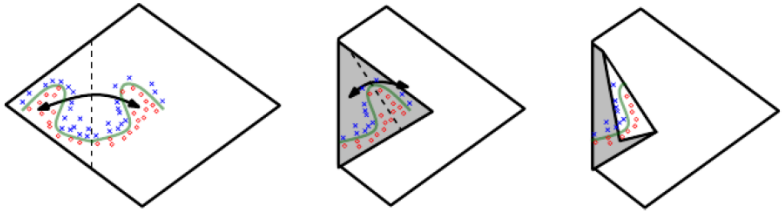
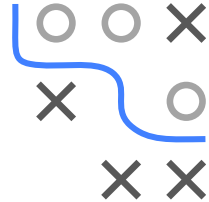


**Figure:** An intuitive, geometric explanation of the exponential advantage of deeper networks formally (Montúfar et al., 2014).

# DEEP NEURAL NETWORKS



Neural networks today can have hundreds of hidden layers. The greater the number of layers, the "deeper" the network. Historically DNNs were very challenging to train and not popular until the late ’00s for several reasons:

- The use of sigmoid activations (e.g., logistic sigmoid and tanh) significantly slowed down training due to a phenomenon known as “vanishing gradients”. The introduction of the ReLU activation largely solved this problem.
- Training DNNs on CPUs was too slow to be practical. Switching over to GPUs cut down training time by more than an order of magnitude.
- When dataset sizes are small, other models (such as SVMs) and techniques (such as feature engineering) often outperform them.

# REFERENCES

Montúfarr, G., Pascanu, R., Cho, K., & Bengio, Y. (2014). *On the Number of Linear Regions of Deep Neural Networks*.